財政學 HW2
繳交期限：2024 年10 月9 日23:59

1. Read the Summary of "The Rationale for Government Intervention" (the supplementary to Lecture I—Introduction 1). (不須繳交)

2. (Rosen 3.6) Imagine a simple economy with only two people: Augustus and Livia.

a. Let the social welfare function be

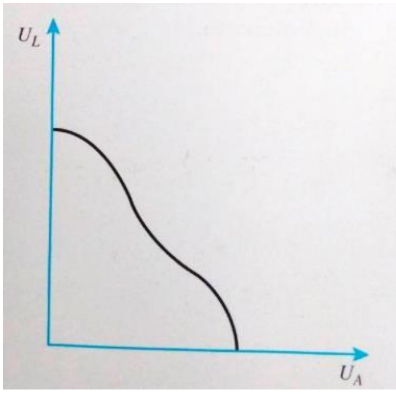
$$W = U_L + U_A$$

where U_L and U_A are the utilities of Livia and Augustus, respectively. Graph the social indifference curves. How would you describe the relative importance assigned to their respective well-being?

b. Repeat part *a* when

$$W = U_L + 2U_A$$

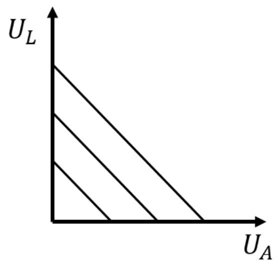
c. Assume that the utility possibilities curve is as follows:



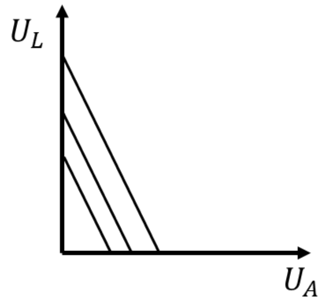
Graphically show how the optimal solution differs between the welfare functions given in parts *a* and *b*.

ANS

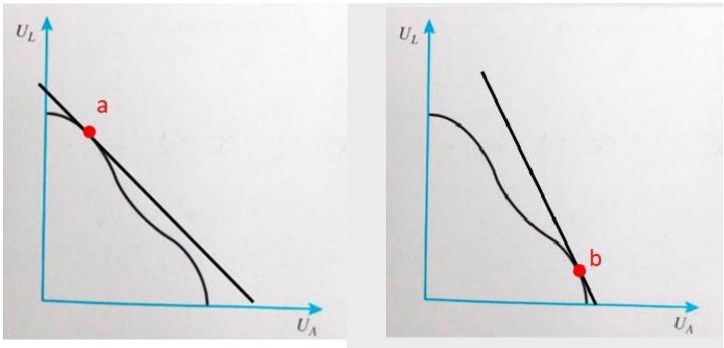
a. Social indifference curves are straight lines with slope of –1. As far as society is concerned, the "util" to Augustus is equivalent to the "util" to Livia.



b. Social indifference curves are straight lines with slope of –2. This reflects the fact that society values a “util” to Augustus twice as much as a “util” to Livia.



c.



3. (Rosen 3.9) Your airplane crashes in the Pacific Ocean. You land on a desert island with one other passenger. A box containing 100 little bags of peanuts also washes up on the island. The peanuts are the only thing to eat.
In this economy with two people, one commodity, and no production, represent the possible allocations in a diagram, and explain why every allocation is Pareto efficient. Is every allocation fair?

ANS

In this case, the “Edgeworth Box” is actually a line because there is only one good on the island. The set of possible allocations is a straight line, 100 units long. Every allocation is Pareto efficient, because the only way to make one person better off is to make another person worse off. There is no theory in the text to help us decide whether an allocation is fair. Although splitting the peanuts even between the people may be fair, it may not be fair if the calorie “needs” of the people are different. With a social welfare function, we can make assessments on whether redistribution for society as a whole is a good thing.

4. (Rosen 3.11) Suppose that Tang and Wilson must split affixed 400 pounds of food between them. Tang's utility function is $U_T = \sqrt{F_1}$ and Wilson's utility function is $U_W = \frac{1}{2}\sqrt{F_2}$, where F_1 and F_2 are pounds of food to Tang and Wilson, respectively.

a. How much utility will Tang and Wilson receive if the food is distributed evenly between them?

b. If the social welfare function is $U_T + U_W$, then what distribution of food between Tang and Wilson maximizes social welfare?

c. If social welfare is maximized when they each obtain the same level of utility, then what is the distribution of food between Tang and Wilson that maximizes social welfare?

ANS

a. If the food is evenly distributed between Tang and Wilson, Tang will have 14.14 units of utility and Wilson will have 7.07 units of utility.

b. If the social welfare function is $U_T + U_W$, then the marginal utilities of both should be equal to maximize social welfare. Equate $MU_T = \frac{1}{2\sqrt{U_T}}$ to $MU_W = \frac{1}{4\sqrt{U_W}}$ and substitute $F_T = 400 - F_W$. Therefore, $F_T = 320$ and $F_W = 80$.

c. If the utility of both Tang and Wilson must be equal, then set $U_T = U_W$ and substitute $F_T = 400 - F_W$ and solve. Therefore, $F_T = 80$ and $F_W = 320$.

5. (Rosen 3.13) Suppose that Hannah's utility function is $U_H = 3T + 4C$ and that Jose's utility function is $U_J = 4T + 3C$, where T is pounds of tea per year and C is pounds of coffee per year. Suppose there are fixed amounts of 28 pounds of coffee per year and 21 pounds of tea per year. Suppose also that the initial allocation is 15 pounds of coffee to Hannah (leaving 13 pounds of coffee to Jose) and 10 pounds of tea to Hannah (leaving 11 pounds of tea to Jose).

a. What do the utility functions say about the marginal rates of substitution of coffee for tea?

b. Draw the Edgeworth Box showing indifference curves and the initial allocation.

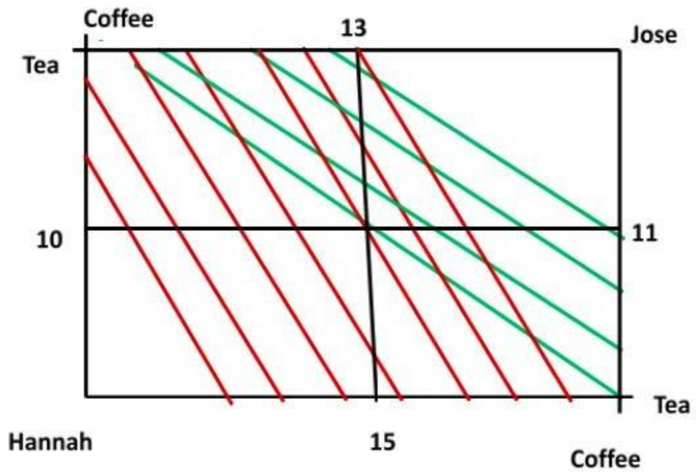
c. Draw the contract curve on the Edgeworth Box. Explain why it looks different from the contract curves depicted in the text.

d. Is the initial allocation of coffee and tea Pareto efficient?

ANS

a. The marginal rates of substitution for coffee and for tea are constant for both Hannah and Jose. Hannah would trade 1/4pound of coffee for 1/3 pound of tea to remain equally satisfied. Similarly, Jose would trade 1/4pound of tea for 1/3 pound of coffee to remain equally satisfied. The constant MRS means linear indifference curves

b. Green indifference curves are Jose's and red indifference curves are Hannah's.



c. The contract curve follows the bottom and right borders of the Edgeworth Box. At any interior point in the box, the parties will find it in their interest to trade until they reach these borders. This is because Hannah would only choose to consume tea once she has consumed all the coffee in the economy. Likewise, Jose would only choose to consume coffee once he has consumed all the tea in the economy.

d. The initial allocation is not Pareto efficient. It is possible to make one better off without making the other worse off.

6. (Rosen 3.14)

Indicate whether each of the following statements is true, false, or uncertain, and justify your answer.

a. If everyone has the same marginal rate of substitution, then the allocation of resources is Pareto efficient.

b. If the allocation of resources is Pareto efficient, then everyone has the same marginal rate of substitution.

c. A policy change increases social welfare if, and only if, it represents a Pareto improvement.

d. A reallocation from a point within the utility possibilities curve to a point on the utility possibilities curve results in a Pareto improvement.

ANS

a. True. In the case of reaching Pareto-efficient points as an interior solution, the Marginal Rate of Substitution (MRS) will be equal.

b. Uncertain. As long as the allocation is an interior solution in the Edgeworth box, the marginal rates of substitution must be equal across individuals. This need not be true, however, at the corners where one consumer has all the goods in the economy.
c. False. A policy that leads to a Pareto improvement results in greater efficiency, but social welfare depends on equity as well as efficiency. A policy that improves efficiency but creates a loss in equity might reduce social welfare.
d. False. Moving to a point on the utility possibilities curve may not result in a Pareto improvement because one party may receive less utility on the curve than they received at the interior point.

7. (Hyman 2.1)
The following table shows how the total social benefit and total social cost of summer outdoor concerts in Central City vary with the number of performances.

Number of Concerts	Total Social Benefit($)	Total Social Cost($)
1	10,000	5,000
2	15,000	11,000
3	18,000	18,000
4	20,000	26,000
5	21,000	36,000

What is the efficient number of concerts?

ANS

The marginal social benefit of the first concert is $10,000, and its marginal social cost is $5,000. The marginal social benefit of a second concert is $5,000, which falls short of its marginal social cost of $6,000. The efficient number of concerts is one. After the first concert, the marginal social cost exceeds the marginal social benefit.

8. (Hyman2.3)
A prominent senator has calculated the total social benefit of the current amount of space exploration at $3 billion per year. The total social cost of space exploration is currently only $2 billion. The senator argues that a net gain to society would result by increasing the amount of space exploration until total costs rise enough to equal total benefits. Is the senator’s logic correct?

ANS

The senator’s logic is false. Equating the marginal social benefit of a service with its marginal social cost maximizes net gains. If output is increased to the point at which TSB = TSC, there will be more than the efficient amount of resources devoted to space exploration.